Assignment 1: Pathfinding in Time-Dependent Railway Networks

Nathan Mekuria Solomon
Student ID: 294533

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1 INTRODUCTION

This assignment investigates pathfinding in a time-dependent railway network constructed from GTFS data [1]. Unlike static graphs, edges are only valid at specific times, which makes time a critical part of the state.

The system supports two objectives: minimizing travel time and minimizing the number of transfers. Additionally, a multi-stop routing problem is solved using Tabu Search.

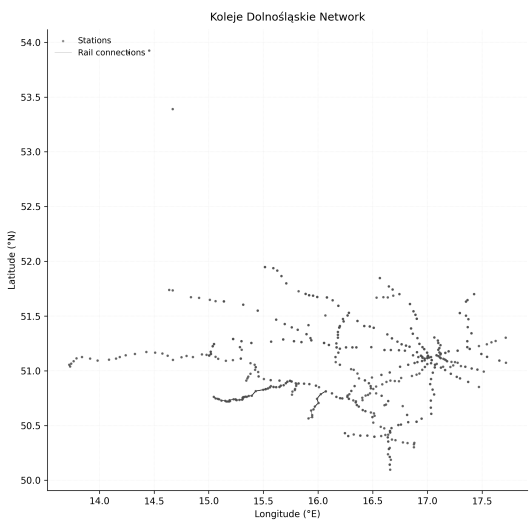

Figure 1: Geographic structure of the railway network derived from GTFS data. Nodes represent stops, and edges represent valid connections between them. This illustrates the structure used for all routing experiments.

2 METHODOLOGY OVERVIEW

All experiments follow a consistent structure. Each query is defined by a start station, a destination, and a departure time. The algorithms are evaluated based on travel time and number of transfers.

A state consists of both a location and a time. This is necessary because connections depend on schedules.

For multi-stop routing, a greedy initial solution is improved using Tabu Search. The algorithm explores neighboring permutations while avoiding recently visited solutions [3].

Each experiment isolates a specific behavior, such as varying departure time or changing the optimization objective.

3 PATHFINDING WITH A*

A* expands fewer nodes than Dijkstra because it uses a heuristic to guide the search toward the goal.

Dijkstra expands nodes uniformly, while A* prioritizes nodes closer to the destination. This reduces unnecessary exploration.

The heuristic estimates the remaining distance to the goal, guiding the search efficiently. This does not affect correctness as long as the heuristic is admissible [2].

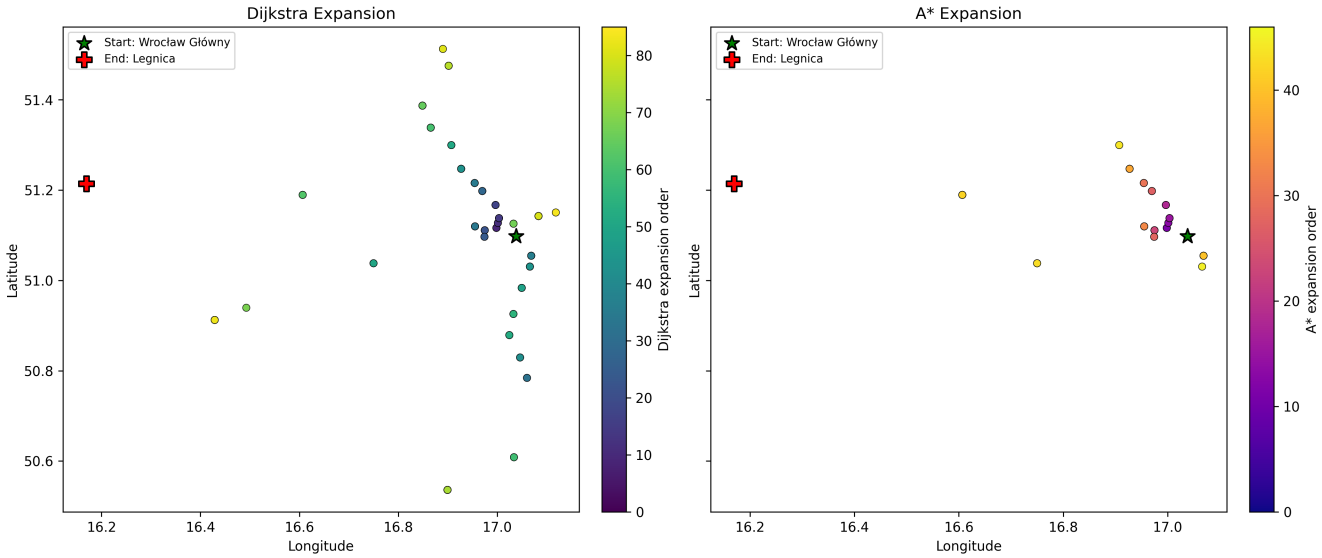

Figure 2: Node expansion comparison for the route from Wrocław Główny to Legnica at 06:00. A* explores fewer nodes due to heuristic guidance, while Dijkstra explores more of the graph.

3.1 Effect of Departure Time

The route from *Wrocław Główny* to *Legnica* was tested across multiple departure times between 05:00 and 09:00.

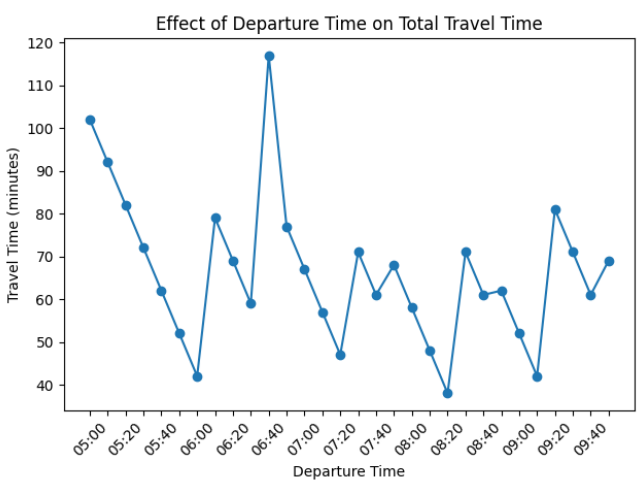

Figure 3: Effect of departure time on total travel time for the route from Wrocław Główny to Legnica. Each point corresponds to a different departure time, showing how schedule changes impact the result.

Small changes in departure time can lead to large differences in travel time. This occurs when a connection is missed, forcing the algorithm to wait or take an alternative route.

3.2 Trade-off Between Objectives

The route from *Wrocław Główny* to *Jelenia Góra* at 06:00 was evaluated using two optimization criteria.

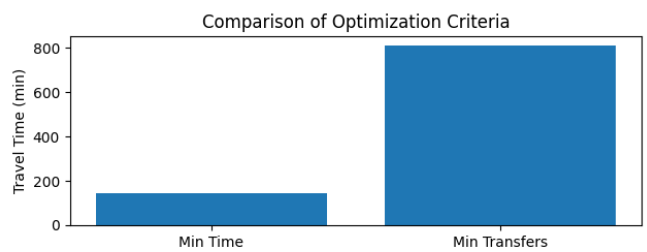

Figure 4: Comparison of optimizing for travel time versus minimizing transfers for the route from Wrocław Główny to Jelenia Góra at 06:00. Both metrics are evaluated to highlight the trade-off between efficiency and convenience.

Minimizing travel time and minimizing transfers lead to different solutions. Faster routes may include more transfers, while fewer transfers often result in longer travel times.

4 TABU SEARCH

A multi-stop route starting and ending at *Wrocław Główny* was optimized using Tabu Search.

The algorithm improves the solution rapidly at first, then stabilizes as fewer improvements remain.

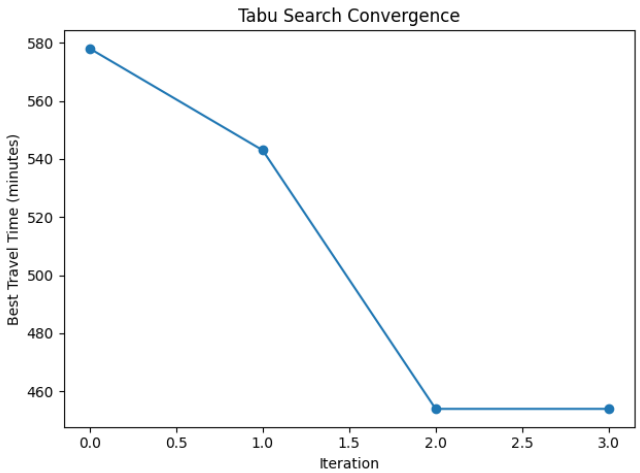

Figure 5: Convergence of Tabu Search for the multi-stop route. The plot shows how the solution improves over iterations before stabilizing.

The tabu list prevents revisiting recent solutions, encouraging exploration.

5 CONCLUSION

The experiments confirm that incorporating time into the state is essential for routing in railway networks. Because connections are only available at specific times, even small changes in departure time can lead to significantly different outcomes in terms of both travel time and waiting periods.

The comparison between A* and Dijkstra demonstrates why heuristics matter in time-dependent search. A* consistently explores fewer nodes because it focuses the search toward the destination using an informed estimate. This reduces unnecessary exploration without sacrificing optimality, provided the heuristic remains admissible.

The results from the objective comparison highlight a fundamental trade-off between travel time and number of transfers. Optimizing for time tends to favor faster but less convenient routes, while minimizing transfers often leads to longer journeys. This reflects a real-world constraint in transportation systems, where efficiency and convenience cannot always be optimized simultaneously.

Finally, the Tabu Search experiment shows that local search methods can effectively improve multi-stop routes, especially in the early iterations. However, the improvements gradually diminish, indicating that the algorithm approaches a local optimum. The use of a tabu list helps prevent cycling and encourages broader exploration of the solution space.

Overall, the experiments show that realistic routing problems require both temporal awareness and careful handling of competing objectives. Simple shortest-path formulations are not sufficient, and more advanced techniques are necessary to model real-world constraints accurately.

AI ASSISTANCE DISCLOSURE

Artificial intelligence tools were used during the development of this work in the following ways:

- Language refinement: improving clarity, grammar, and readability.
- Code documentation: assisting in writing and refining comments.
- Debugging support: helping identify and resolve logical and syntactical issues.

All design decisions, implementation, and final content were reviewed and finalized by the author.

References

[1] Google Developers, *GTFS Schedule Reference*, 2024.

[2] P. Hart, N. Nilsson, B. Raphael, *A Formal Basis for the Heuristic Determination of Minimum Cost Paths*, IEEE, 1968.

[3] F. Glover, *Tabu Search—Part I*, ORSA Journal on Computing, 1989.